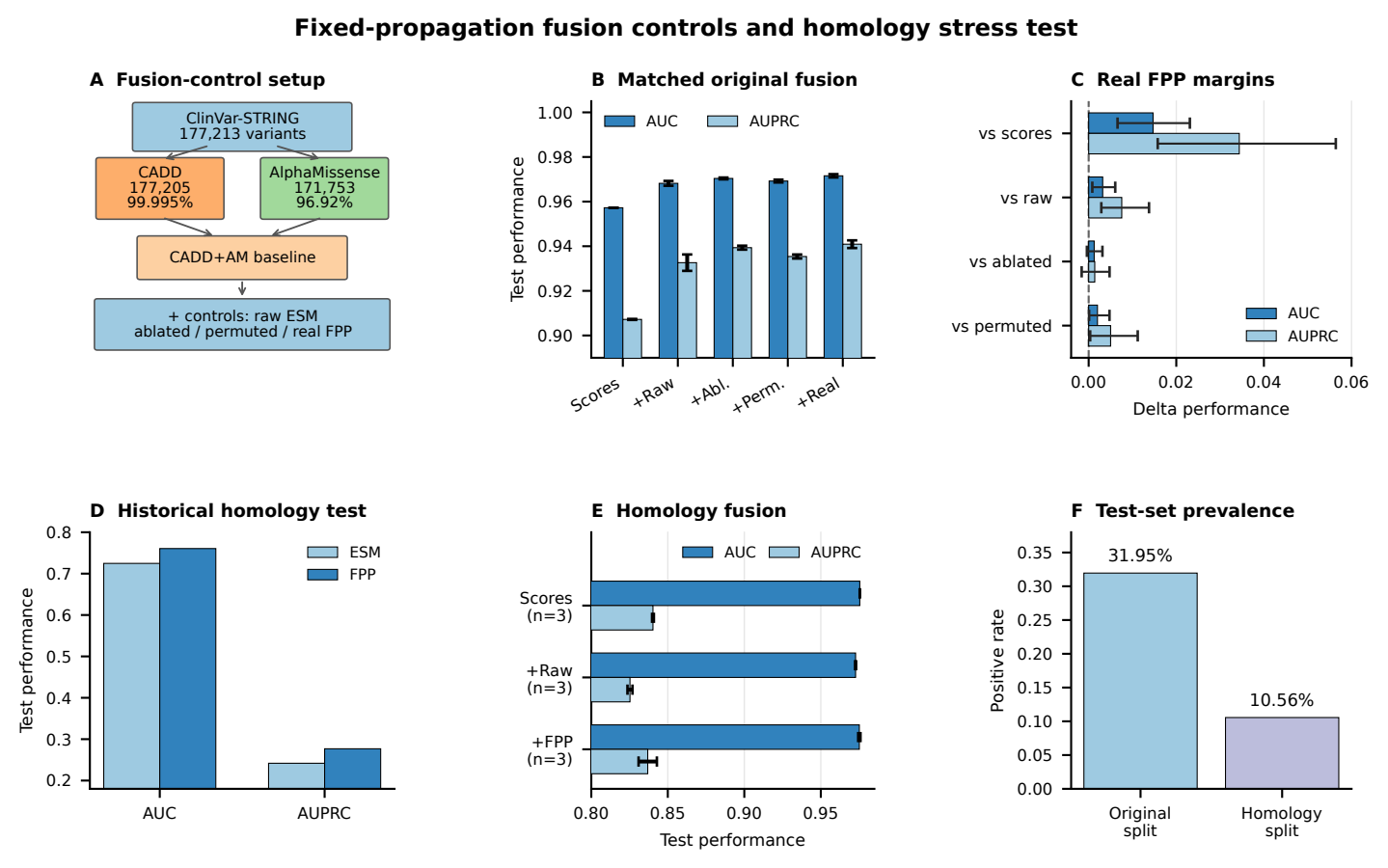
Fixed-propagation fusion controls and homology stress test
A Fusion-control setup
ClinVar-STRING
177,213 variants
CADD
177,205
99.995%
AlphaMissense
171,753
96.92%
CADD+AM baseline
+ controls: raw ESM
ablated / permuted / real FPP
B Matched original fusion
AUC
AUPRC
Test performance
1.00
0.98
0.96
0.94
0.92
0.90
Scores
+Raw
+Abl.
+Perm.
+Real
C Real FPP margins
vs scores
vs raw
vs ablated
vs permuted
AUC
AUPRC
0.00
0.02
0.04
0.06
Delta performance
D Historical homology test
ESM
FPP
Test performance
0.8
0.7
0.6
0.5
0.4
0.3
0.2
AUC
AUPRC
E Homology fusion
AUC
AUPRC
Scores
(n=3)
+Raw
(n=3)
+FPP
(n=3)
0.80
0.85
0.90
0.95
Test performance
F Test-set prevalence
31.95%
10.56%
Positive rate
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Original
split
Homology
split